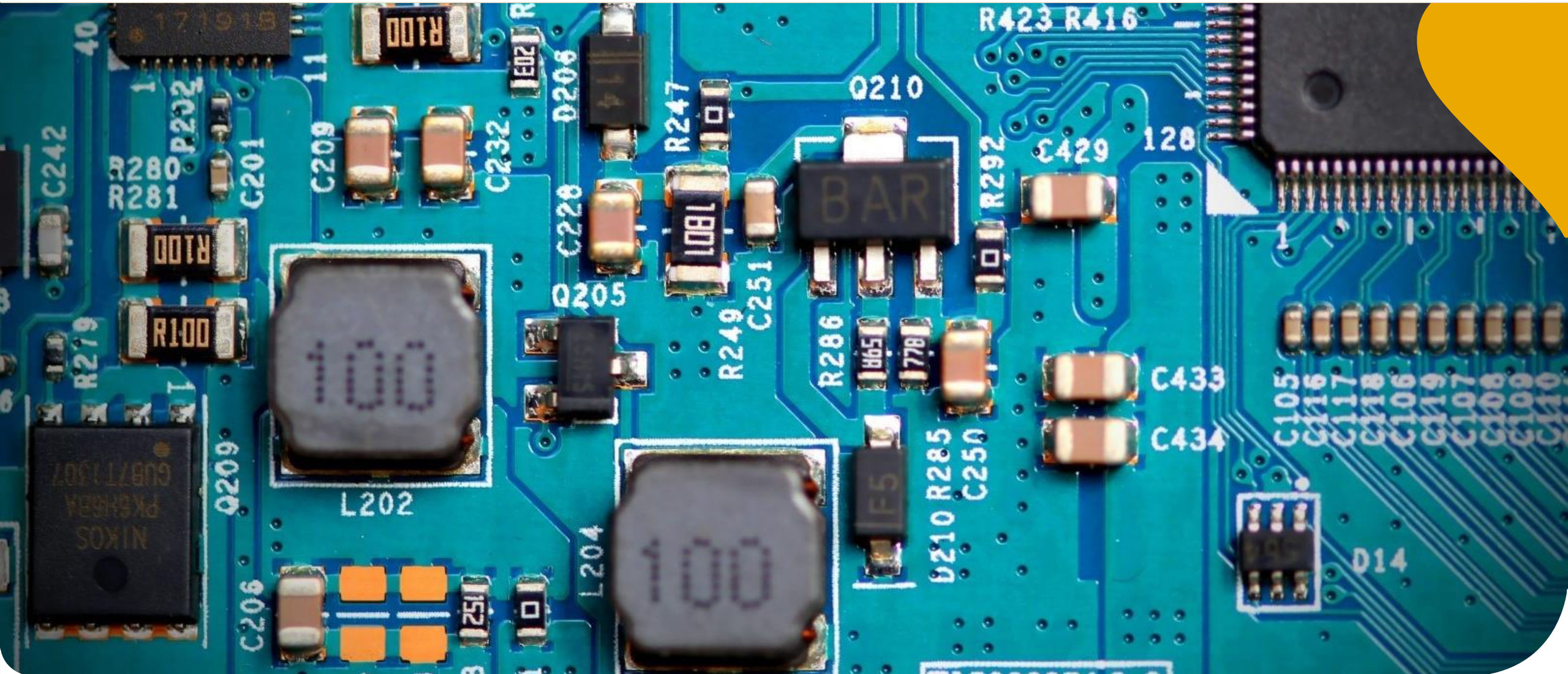


Design(E)-314

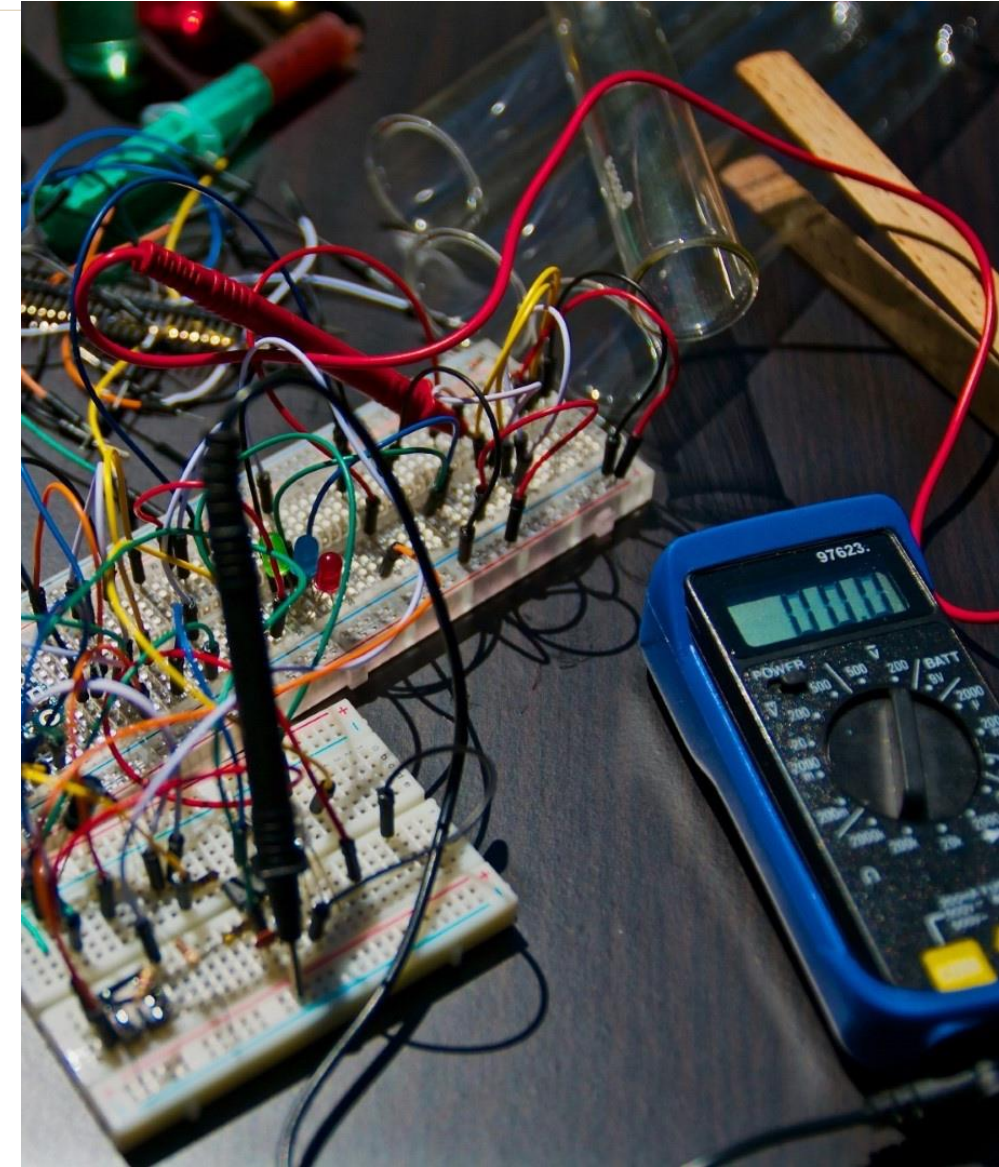
Lecture 4



Overview

We are in the Demo 2 cycle

- Test 1 – Review
- Ultrasonic sensor
- Demo 2
- GIT
- Test 2
- Demo 3 cycle
 - Light sensor
 - Op-Amp
 - ADC



- Version 0.5 is the latest version

Version	Date	Changes
0.1	9 Feb 2026	- Initial project specification and guide
0.2	11 Feb 2026	- Added Demo 1 information in Tables 9 and 10
0.3	19 Feb 2026	- Clearly specified the Demonstration 1 UART message format for the temperature sensor in Tables 9.
0.4	3 Mar 2026	- Added Demo 2 information in Tables 9 and 10 - Added information related to preliminary report
0.5	5 Mar 2026	- Added clarification to checking of alarm conditions, and SB response to Stat command.

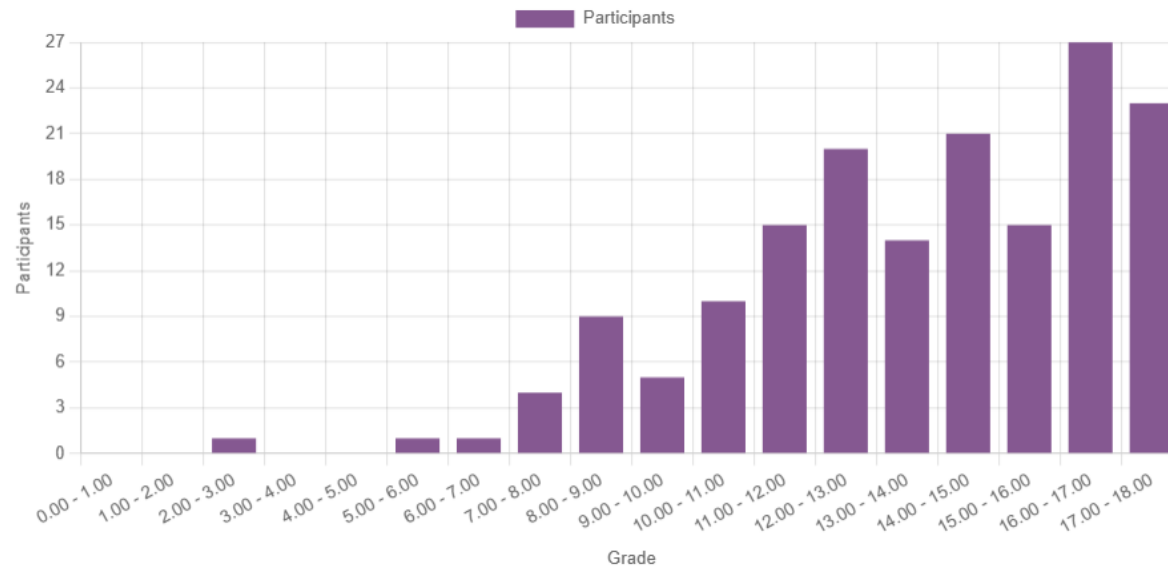
- Major change in v0.5: Specification for Demo 2 and preliminary report

Test 1

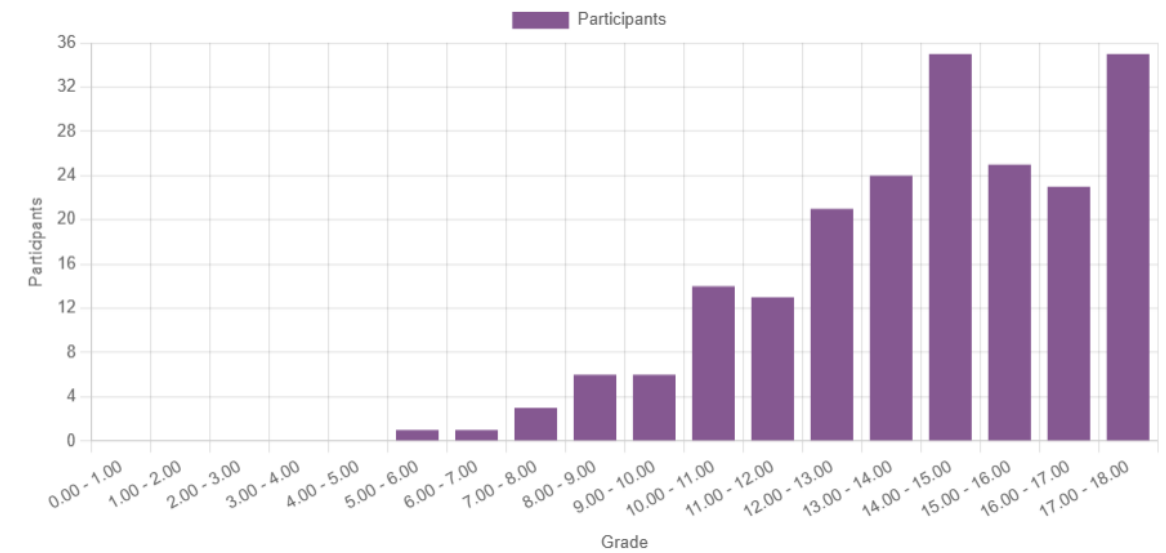
Review and Feedback

Results

Overall number of students achieving grade ranges



Overall number of students achieving grade ranges



Test 1

Review and Feedback

UART Question

- A UART port is configured to operate at 57600 baud, with 8-bit data, one start bit and one stop bit, and even parity bit. What is the maximum data transfer rate on the channel, in bytes per second (rounded to a full byte)?
- Answer:  5236 (Total bits = 11 required to send a single byte)

Test 1

Review and Feedback

Oscilloscope Question

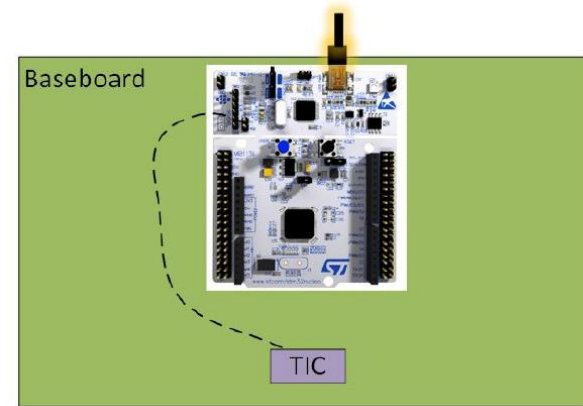
- Consider the following set of oscilloscope readings from measuring sent UART data:
- **Answer:** 5236 (Total bits = 11 required to send a single byte, not 8)
 - * Default start bit → Always low
 - * Default stop bit → Always high

MCU vs LMT01 clock speed

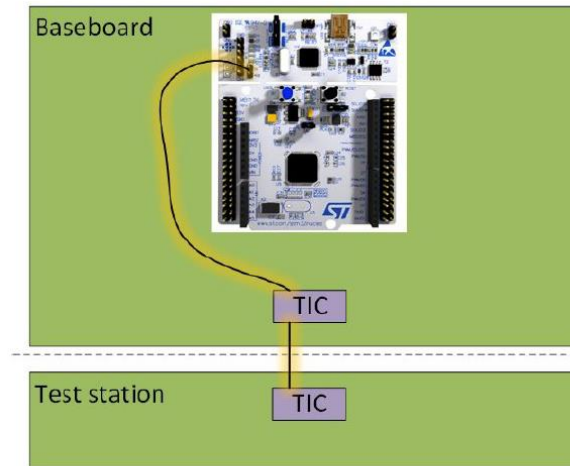
- An STM32 MCU runs at 84 MHz.
- An LMT01 sensor outputs pulses at 88 kHz.
- The LMT01 is connected to a GPIO pin as an external interrupt on the STM32
- How many full MCU clock cycles occur during one LMT01 sensor pulse period?
- **Answer:** 954 full cycles

TIC Connections

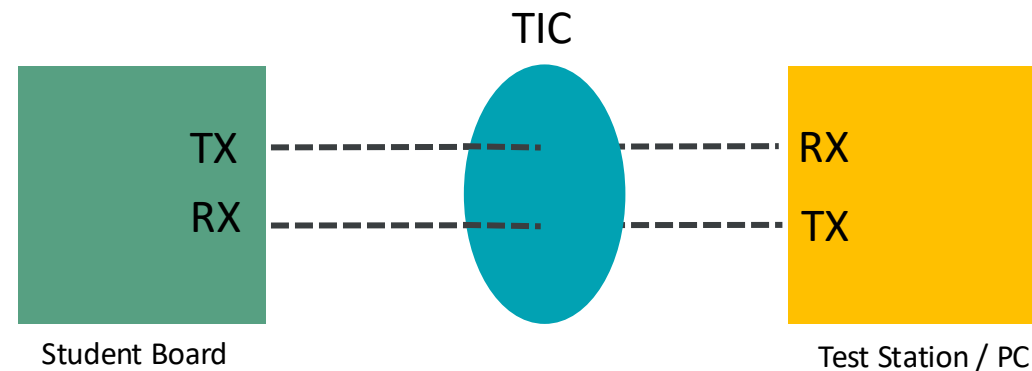
UART



Student performs test or debugging without test station. UART appears as a Virtual COM port on PC



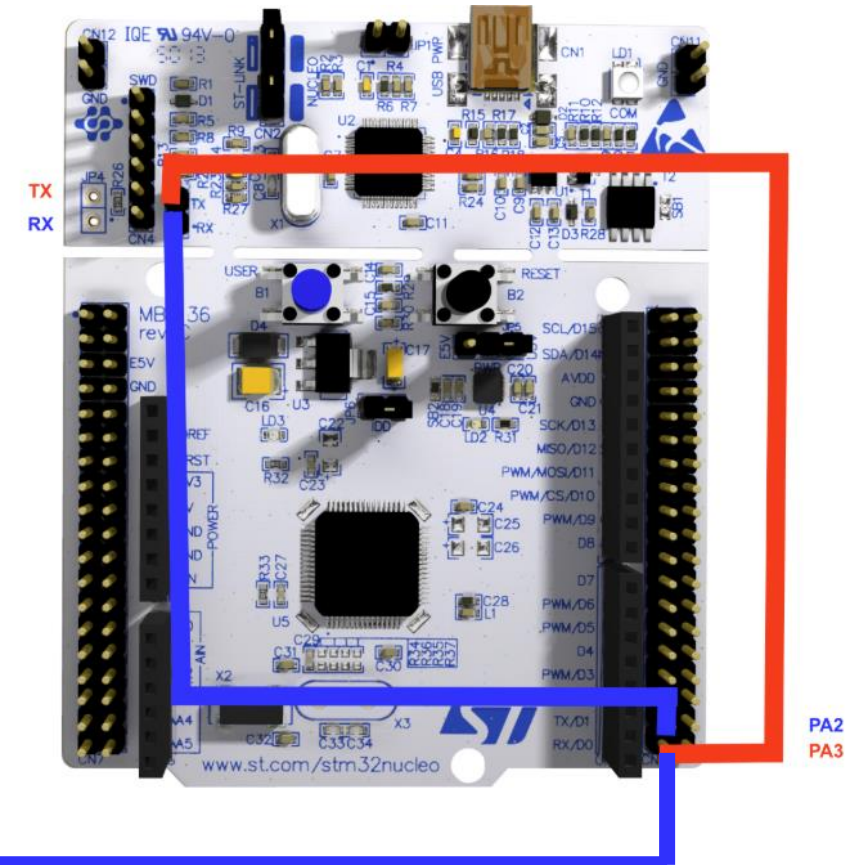
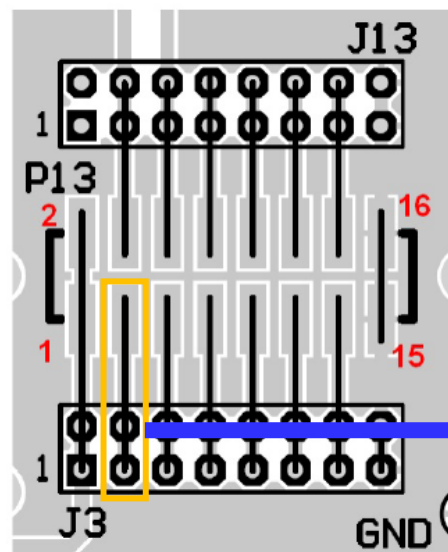
Automated testing through Test Station - UART is routed to TIC



TIC Connections

UART

P13 pin	J3/J13 solder connection	Signal
1	J3 - 1,2	V_bat - 9V Supply from test station
2	N/C	V_bat - 9V Supply from test station
3	J3 - 3,4	UART transmit (TX from student board, RX of test station)
4	J13 - 3,4	5V supply from student board (fed back to test station for verification/measurement)
5	J3 - 5,6	UART receive (RX from student board, TX of test station)
6	J13 - 5,6	3.3V supply from student board (fed back to test station for verification/measurement)
7	J3 - 7,8	Button UP-S1 (connected to TS for debug/verification)



LMT01

Temperature sensor

Implementation/Timing:

- You can use either EXTI **or** TIMER (external clock mode) to count pulses...
- We know that there are about x954 MCU clock (84MHz) cycles between each 88kHz pulse. In real life how would I know how many ISR (interrupt service routine) cycles my LMT01 sensor actually requires? * NB
- ISR cost is determined by compiler output + MCU architecture + what you put inside the ISR.

Method 1:

Use the Cortex-M DWT (Data Watchpoint and Trace) Cycle Counter

- Procedure

1. Enable DWT
2. Reset cycle counter
3. Read counter at ISR entry
4. Read counter at ISR exit
5. Subtract

$(\text{ISR exit} - \text{ISR entry})$

pulse → time must be calculated

Method 2:

GPIO Toggle + Oscilloscope

Procedure

1. GPIO-PIN = HIGH;
2. ... ISR code ...
3. GPIO-PIN = LOW;

- Then measure pulse width on a scope.
- Convert time to cycles using: $\text{cycles} = \text{time}[s] \times \text{fclock}[Hz]$
- **Example:** If time = 1.2us, cycles = 1.2us x 48MHz = 101 cycles

If your ISR is:

80 cycles → trivial (≈8% CPU)

300 cycles → noticeable (≈31%)

900 cycles → catastrophic



Pro-Tip:

Go to your .ioc file and select “Clock Configuration” to make sure which clock speed you are running at.

HC-SR04 Ultrasonic Sensor

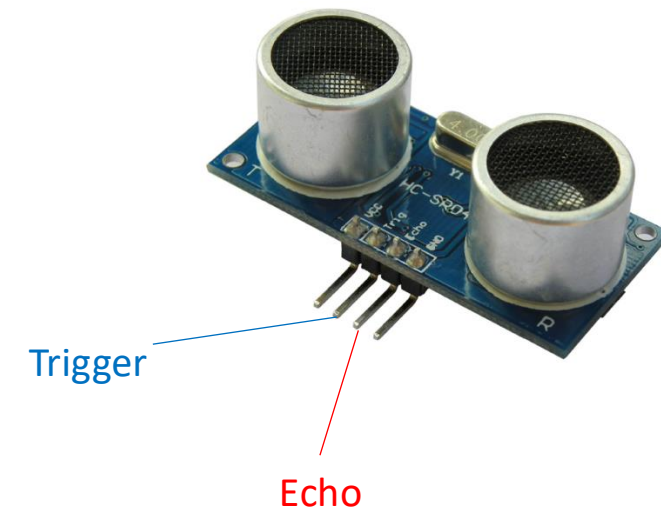
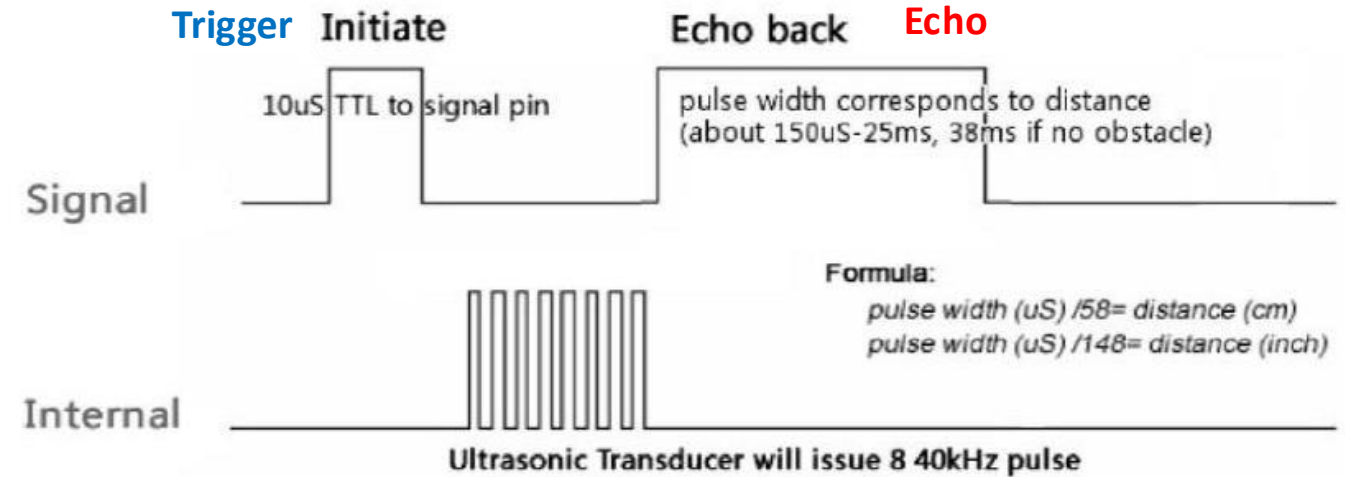
Basic operation



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1. Start measurement with high (5V) pulse for 10us,
2. HC-SR04 activates and sends 8x cycle ultrasonic burst, of 40 kHz, which will travel at the speed of sound.
3. Echo listens for wave to be reflected from an object
4. Echo pin will time-out after 38ms (i.e. no object detected), but triggers before 38ms if object detected



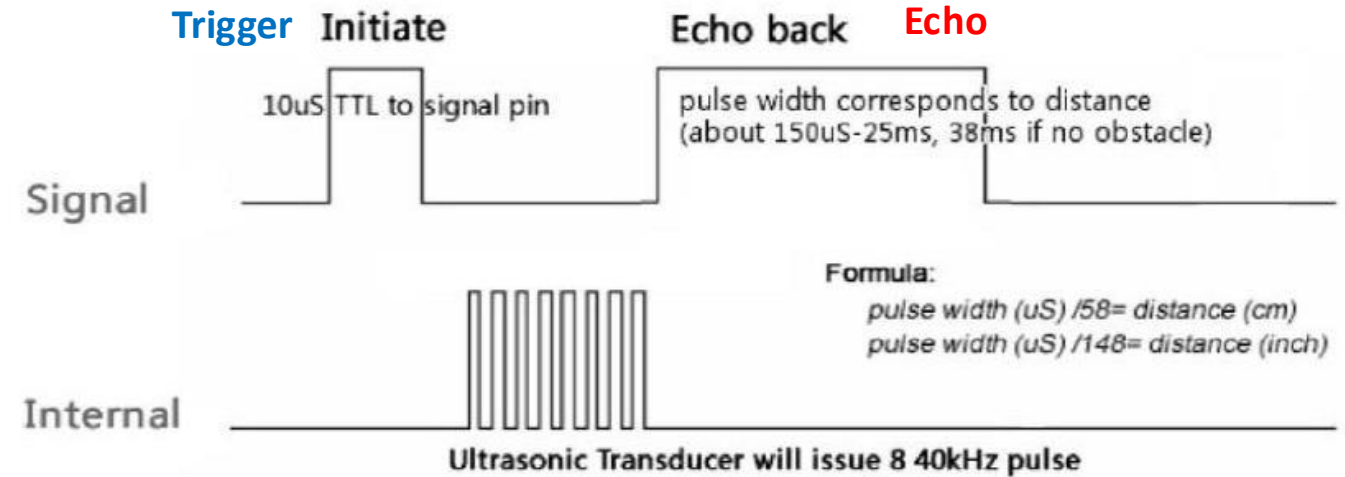
HC-SR04 Ultrasonic Sensor Hardware



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- Have you confirmed ECHO pin output voltage?
 - 3.3V or 5V
- Is it necessary to supply the device with 5V?
- How will you activate the Trigger pin if 5V is required?
- How will you read the Echo pin if the output is 5V (and not 3.3V)



HC-SR04 Ultrasonic Sensor Hardware



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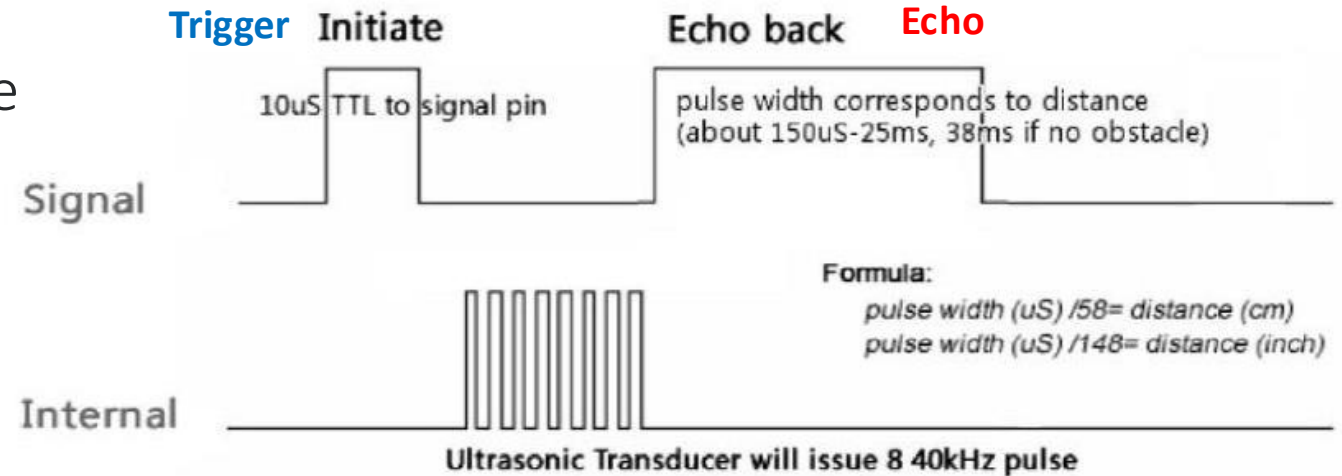
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- Can you use an interrupt pin to read the Echo pulse?

- Difference between time at interrupt rising edge and time and falling edge is the pulse width

- If you use `uwTick()` or `HAL_GetTick()`, what is the max resolution of the pulse width received from the echo line?

- Sound travels at 343m/s , which means the maximum accuracy/resolution of your dist. measurement depends on the time resolution of `HAL_GetTick()`



HC-SR04 Ultrasonic Sensor Software



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- Consider the use of a TIMER to increase your timing resolution

↳ We used a timer

Echo back

pulse width corresponds to distance
(about 150uS-25ms, 38ms if no obstacle)

Echo

- Option 1: Configure a timer running at a known frequency. When an edge arrives, the timer copies its current value into a capture register.

- $\Delta t = t_{falling} - t_{rising}$

- Use channel 1 and channel 2
 - Channel 1 = rising, Channel 2 = falling

* These were the settings I used

Pro-Tip:

Set prescaler so the timer increments every $1\ \mu\text{s}$.

PSC = 84 - 1 (for 84MHz clock)

Prescaler = 83
Counter mode = Up
Period = 0xFFFFFFFF
Clock division = 1



HC-SR04 Ultrasonic Sensor Software



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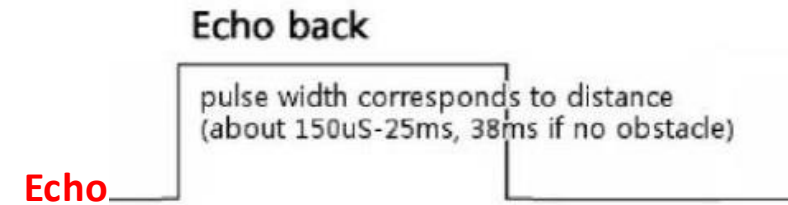
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Option 2: Use TIMER Gated Mode.

Core idea: Timer Slave Mode with Gated Counting:

- Only counts while the input signal is HIGH. When the pulse goes LOW, the counter automatically stops.

Note: Set Trigger Source as TI1FP1



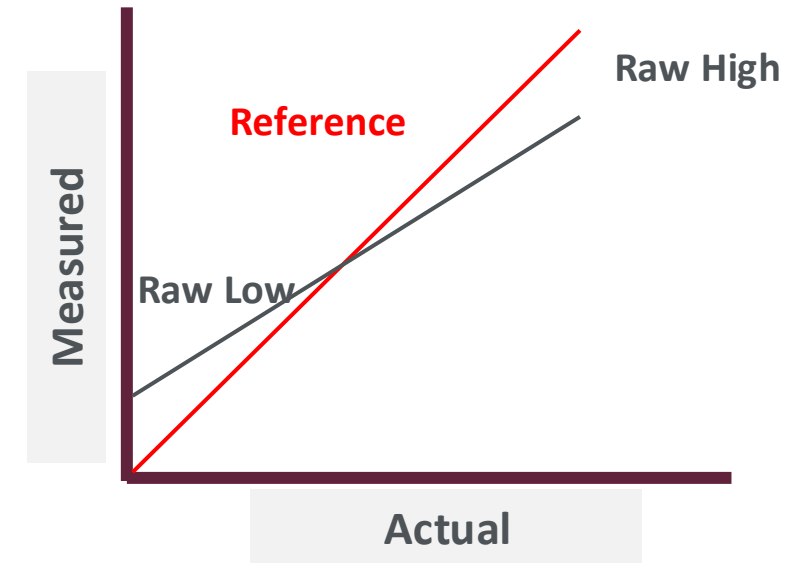
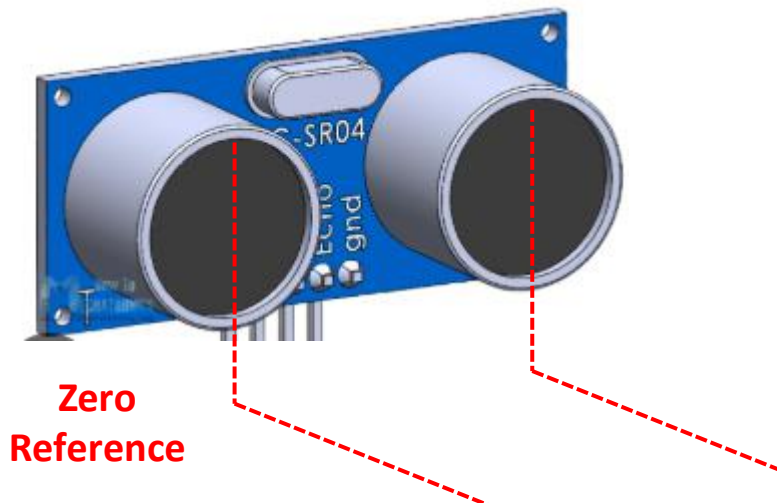
Pro-Tip:

Your time variables should always be `uint32_t`

HC-SR04 Ultrasonic Sensor Calibration

Calibration/Confirmation:

- To ensure you are reading the right value, you must calibrate/investigate relative to a known distance value. ***NB**
- **Note:** Your reference point for calibration is at the edge of the ultrasonic sensor.



Demo 2 requirements

What to expect

2	From Demo 1 On power-up, the TS will expect your student number via UART using the prescribed message format, no sooner than 100ms after power-up and no later than 500ms after power-up. The TS will simulate button pushes with and without bounce for push-buttons S1-S5. The TS will override the tactile push-buttons (via TIC connector P13 pins 7, 8, 9, 10, 11) while monitoring the status-indicating LEDs to check for correct system operation. The maximum response time after button pushes is 500 ms. The TS will monitor and test the 3.3V and 5V lines. Additionally: The TS will send <i>Stat</i> telemetry requests at different intervals via UART to the SB. The SB responses will be monitored for correct system functioning. The minimum delay between consecutive UART commands will be 500 ms. The system should update and respond IMMEDIATELY (<100 ms) to any UART input. The SB must be able to provide the measured ambient temperature and the measured distance of an object relative to the ultrasonic sensor. The TS will monitor the status-indicating LEDs and monitor whether an uncomfortable temperature condition and/or a proximity warning condition is reached. The SB must switch on the correct status-indicating LEDs to indicate the alarm condition. The appropriate LED must flash at the correct frequency. GIT repos will be checked, and it must be possible to successfully compile your software (from your cloned repository) into a binary file. It must also be possible to program your microprocessor using the just compiled binary file.	From Demo 1: SS1, SS2 (student number), SS6 (Temperature), SS7 (LEDs), SS8 (tactile push-buttons), SS12 (5V, 3.3V, UART-RX, UART-TX). Additionally: SS2 (Stat telemetry request), SS3 (ultrasonic sensor), SS7 (alarm condition indicators).
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Demo 2

Marks allocation



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4x Parts:

- Test 0: GIT [4]
- Test 1: Voltages [2]
- Test 2: Start up [1]
- Test 3: Button command [5]
- Test 4: UART Stat command [5]
- Test 5: Temperature sensor [3]
- Test 6: Ultrasonic sensor [5]



Note:

5% and 10% tolerances are checked

Alarm conditions checked with corresponding LED behaviour

Various ranges of values will be checked for temperature and distance

Demo 2

GIT Marks

GIT = 4 marks:

Test 0: GIT check [4]

- Multiple commits [1]
- Regular use of Git confirmed [1]
- Can compile from cloned repository [1]
- Program MCU from latest commit [1]

* Very important to add debug directory



Commit to test Demo2 test stations
Herman Engelbrecht authored 3 days ago

ab48d794 

main ▾ 12704482-e314-proj / + ▾

Find file Web IDE ▾  ▾ Clone ▾

 README

 Add LICENSE

 Add CHANGELOG

 Add CONTRIBUTING

 Set up CI/CD

Name	Last commit	Last update
 .settings	Initial commit	2 weeks ago
 Core	Commit to test Demo2 test stations	3 days ago
 Debug	Commit to test Demo2 test stations	3 days ago
 Drivers	Initial commit	2 weeks ago
 FATFS	Initial commit	2 weeks ago
 Middlewares/Third_Party/FatF...	Initial commit	2 weeks ago
 Release	Commit just after Demo 1	1 week ago
 .cproject	Initial commit	2 weeks ago
 .gitignore	Work done until demo 1	1 week ago
 .mxproject	Initial commit	2 weeks ago
 .project	Initial commit	2 weeks ago
 E-DAS Debug.launch	Work done until demo 1	1 week ago
 E-DAS.ioc	Work done until demo 1	1 week ago
 LEDFlash.launch	Initial commit	2 weeks ago
 README.md	Initial commit	2 weeks ago
 STM32F411RETX_FLASH.ld	Initial commit	2 weeks ago
 STM32F411RETX_RAM.ld	Initial commit	2 weeks ago

Demo 2 requirements

Demo Rules

You must be ready to Demo at the start of the Demo. However, since we cannot Demo everyone in parallel, we have a 3-hour time slot.

Rules

- **1st Demo opportunity:**
 - Must occur within 2 hours:
 - You must be in the line by latest 2 hours after the start of the Demo
 - 2nd Demo opportunities: Will be bypassed 2 hours after the start of the Demo, to ensure everyone has access to at least 1x Demo opportunity.
- **2nd Demo opportunities:**
 - Line closes 30 minutes before end of Demo session.

Content

- All Demo 1 & 2 requirements are subjects for the test
- Test is there to ensure that you know what you are actually doing
- Hardware, Software, Theory questions will be asked
- “Open book” i.e. bring your lecture, code & design notes to the test
- No Google/Internet connection allowed

Test 2



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How to prepare

- Write up on all of your hardware and software design elements that were required for Demo 1 (Content/Design unrelated to Demo 1 will not serve as test content)
- Expect: Data sheet related questions
- Expect: Hardware design questions
- Expect: Software design/code implementation questions
- Expect: Results/Interpretation questions



Pro-Tip:

Bring a pen and paper to the test

Bring a calculator to the test (much faster than PC calculator)

Test 2

Details:

- Tuesday 17 March
- Venue: M2004, M2002 and/or M3002
- You must write the test in your allocated lecture time slot
- If you write the test in another time slot, it will simply be regarded as an “absent” result (i.e. 0%)
 - M&M: 08:10 – 09:00 AM
 - E&E, Comp. Sci: 09:10 – 10:00 AM



Pro-Tip:

Arrive 10 minutes early, log into your PC, and complete 2-factor authentication if required.

No cellphones allowed during test

Reporting

Take Note

- When you take measurements for your report, your Student Card must be visible in the photos
- This provides proof to ECSA that these are your own and unique measurements.

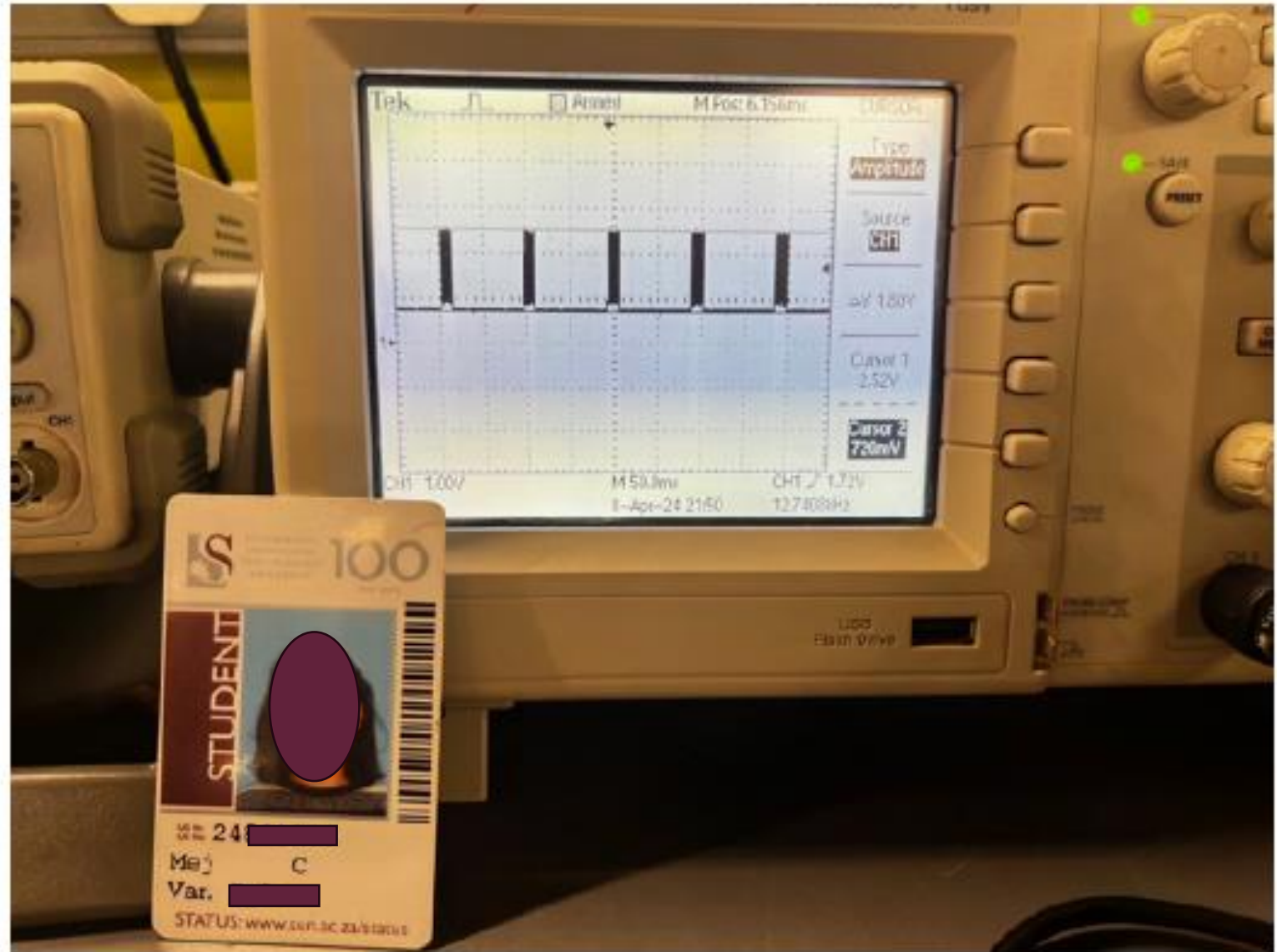


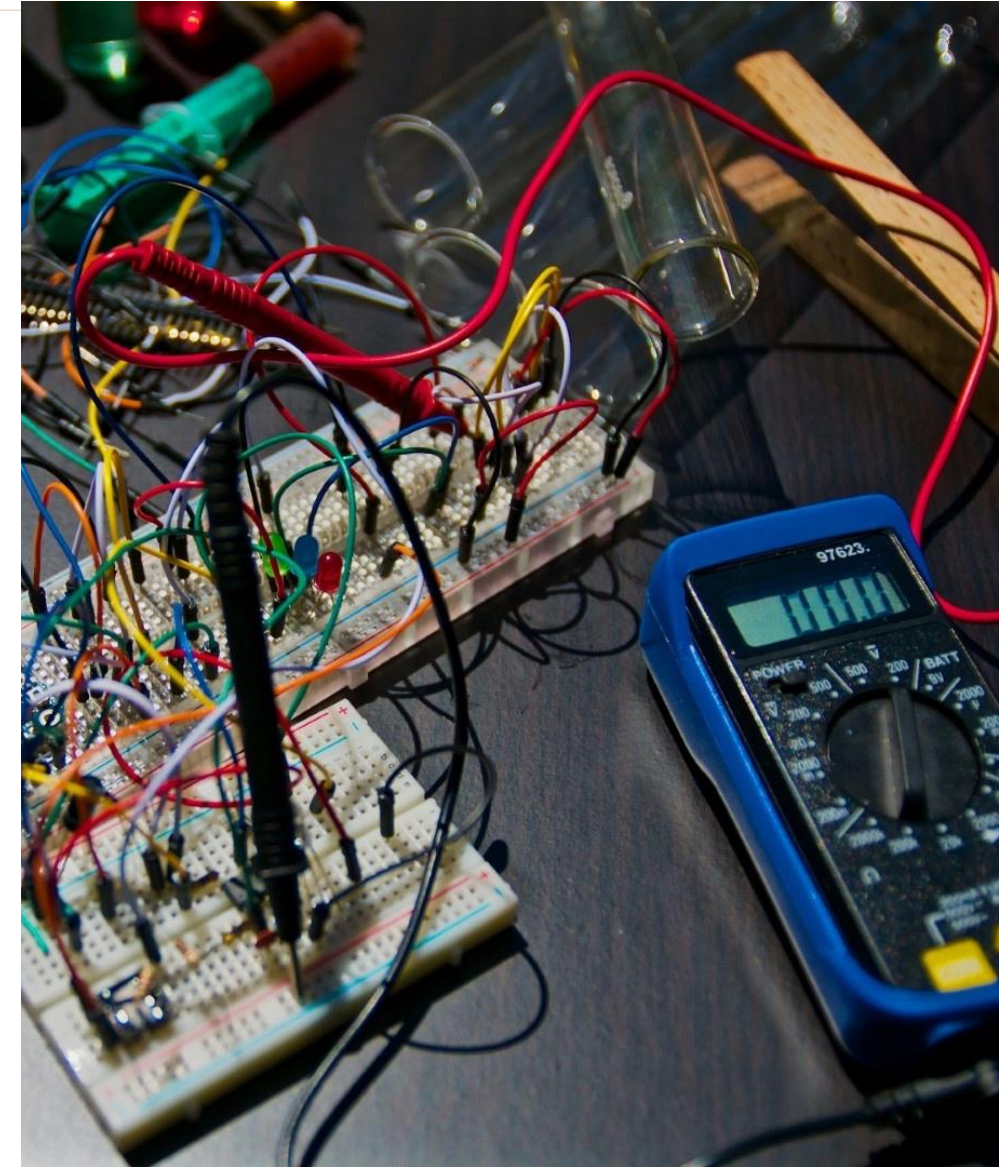
Figure 26: Oscilloscope reading to measure sensor pulses.

Demo 3 Cycle

The following content is for the Demo 3 cycle

- Light sensor,
- Op-Amp,
- ADC concepts

→ In order to implement light sensor
ADC must be applied



Demo 3 Cycle



- Light sensor,
- Op-Amp,
- ADC concepts

Light Sensor (Photo Diode)

SFH203



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- The objective of the light sensor is to inform the driver of low light conditions (i.e. turn on head lights)
- Drivers should turn on their headlights when ambient light levels drop below 300 lux, which generally occurs during dusk, dawn, or in tunnels.
- Lab conditions are about 1000 – 2000 Lux
- Sunshine is about 30,000 – 100,000 Lux



Light Sensor (Photo Diode)

SFH203

Operation

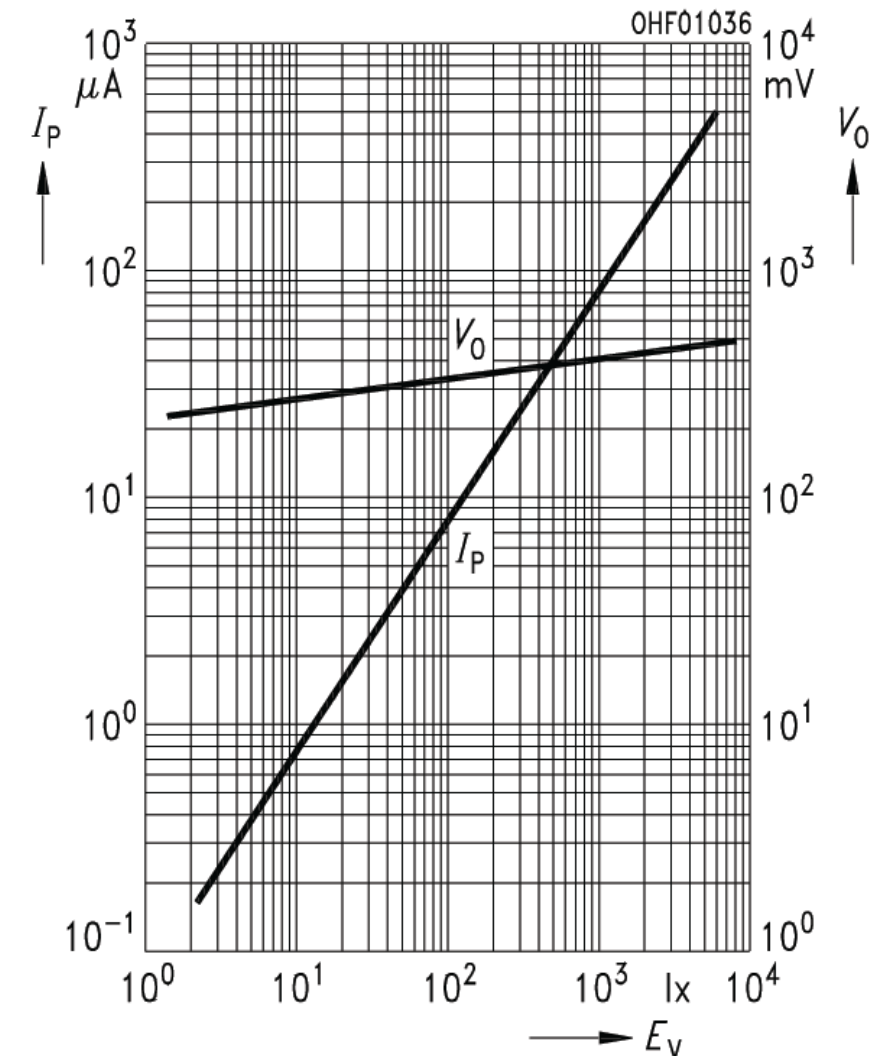
Acts like a current source

- Photodiode output current is directly proportional to light intensity (Lux)
- Design should consider for output current during regular sunshine which can be 30,000 – 100,000 Lux, but should be sensitive enough for low light conditions

Light sensor conversion

* Objective: How bright out is it
Do I need to turn on my headlights

- Formula to convert Output Current >> LUX
- Datasheet graph is in log-scale on x and y axis, with linear relationship
- Assume linear relationship $y = mx + c$



Light Sensor Operational Amplifier



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Photo diode outputs current
STM can't read current

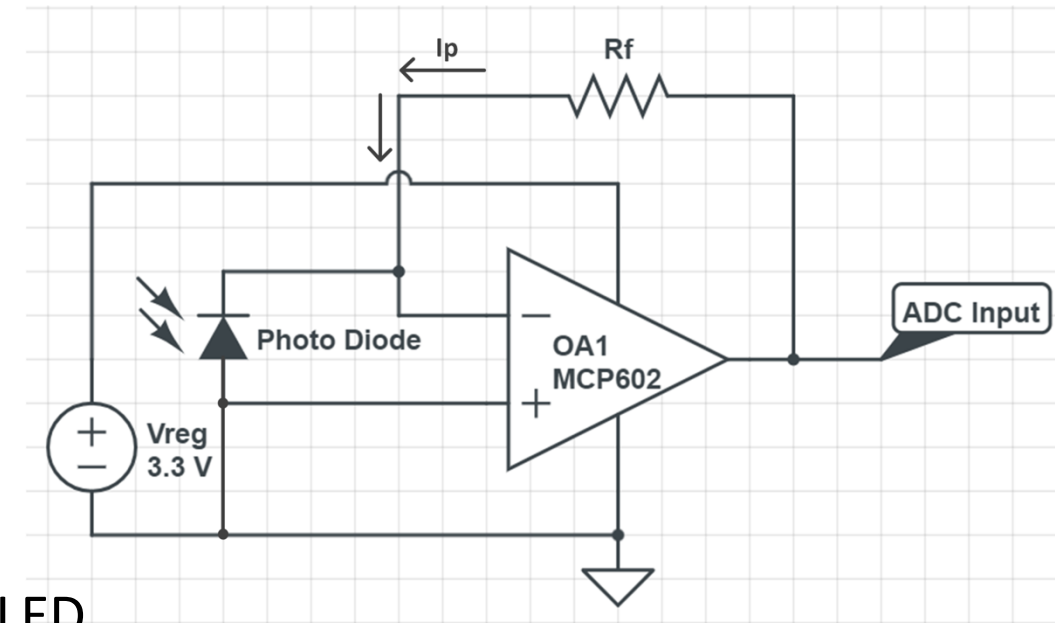
- How do we translate current (A) into a voltage (V)?
- How does voltage correlate to LUX?

Op-Amp Design

- Use MCP602 Op-Amp for your circuit design
- Op-Amp Design provided

Calibration

- You must calibrate your photo-diode using the White LED that was provided to you.
- Ensure a high level of accuracy 0 – 1000 Lux
- Design the board to provide 3.3V at a maximum of 1000 LUX
- Allows you to take advantage of the full ADC resolution (0 – 3.3V)



Use this circuit

Do it in such a way that
0-3.3V ↔ 0-1000

ADC conversion

ADC measurement

Op-Amp circuit supply voltage?
5V vs. 3.3V

ADC: 0 – 3.6V (max) explained
Should we design for 3.6V?

ADC resolution
 $\leq 5\%$ accuracy expected

**NB*

Pro-Tip:

- Do not design your ADC for the maximum input range to avoid 'clipping'.



ADC measurement

- The raw ADC value is obtained using HAL library functions
- ADC returns a 12 bit value.
- Since $2^{12} = 4096$ the range is 0 - 4095, where zero indicates 0V and 4095 indicates the reference voltage.
- ADC output needs to be converted to a voltage
- The measured ADC voltage needs to be calibrated



Pro-Tip:

- Calibrate your ADC voltage measured for a range of 3-5 values between 0V and Vref

Demo 2 Cycle Preliminary Report



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Preliminary report

- Submission closes: **7 April 2026 @ 12:00** (mid-day)

Peer-review process

- Each student will peer-review 3x other reports
- Use the rubric provided to assess the designs.
- Provide sensible feedback to your peers regarding their report / designs and what they can do to improve their report / designs in future.
- Submission closes: **13 April 2026 @ 12:00** (mid-day)

Marking/Weights:

- 50%*obtained mark + 50%*peer-review-scoring-ability



Pro-Tip:

- Follow the rubric to the best of your ability
- Your ability to review a design weighs equal to your ability to communicate your design

Questions?

